

LINUX SYSTEM ADMINISTRATION

Comprehensive Notes & LFCS Exam Reference

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LFCS EXAM ALIGNED

DISTRO-AGNOSTIC

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Chapter 1

User & Group Management

Identity, Authentication & Privilege

Linux uses a discretionary access control model where every process and file is owned by a user and a group. Understanding user and group management is fundamental to system security.

1.1 /etc/passwd — The User Database

Each line in /etc/passwd has seven colon-delimited fields:

```
username:password:UID:GID:GECOS:home_dir:shell
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
alice:x:1001:1001:Alice Smith:/home/alice:/bin/bash
```

- password field 'x' means the hash is stored in /etc/shadow
- UID 0 = root; UIDs 1–999 = system accounts; 1000+ = regular users (on most distros)
- shell /sbin/nologin or /bin/false prevents interactive login

1.2 /etc/shadow — Password Hashes

Readable only by root. Contains hashed passwords and aging information:

```
username:hashed_pw:lastchg:min:max:warn:inactive:expire:reserved
alice:$6$salt$hash...:19000:0:90:7:::
```

- Field 3 (lastchg): days since Jan 1 1970 the password was last changed
- Field 5 (max): maximum number of days between password changes
- Hash prefix \$6\$ = SHA-512, \$5\$ = SHA-256, \$1\$ = MD5 (avoid)

1.3 Managing Users

Command / Syntax	Description
useradd username	Create user with defaults
useradd -m -s /bin/bash -c 'Full Name' -G create -d /home	Create user with home dir, shell, comment, group
useradd -r -s /sbin/nologin svcuser	Create system account (no login)
usermod -aG wheel alice	Append alice to wheel group
usermod -l newname oldname	Rename a user
usermod -L alice	Lock account (prefix ! in shadow)
usermod -U alice	Unlock account
usermod -e 2025-12-31 alice	Set account expiry date
userdel alice	Delete user (keep home dir)
userdel -r alice	Delete user AND home directory
passwd alice	Set/change user password
passwd -l alice	Lock password
passwd -u alice	Unlock password

Command / Syntax	Description
<code>passwd -e alice</code>	Force password change on next login
<code>chage -l alice</code>	List password aging info
<code>chage -M 90 -W 7 alice</code>	Set max age 90 days, warn 7 days
<code>id alice</code>	Show UID, GID and all groups
<code>whoami</code>	Show current effective username
<code>w / who</code>	Show logged-in users
<code>last</code>	Login history from <code>/var/log/wtmp</code>

1.4 Managing Groups

Group information is stored in `/etc/group` (and hashes in `/etc/gshadow`):

```
groupname:password:GID:member1,member2
```

Command / Syntax	Description
<code>groupadd devops</code>	Create group devops
<code>groupadd -g 2000 devops</code>	Create group with specific GID
<code>groupmod -n newgrp oldgrp</code>	Rename group
<code>groupdel devops</code>	Delete group
<code>gpasswd -a alice devops</code>	Add alice to devops
<code>gpasswd -d alice devops</code>	Remove alice from devops
<code>gpasswd -A alice devops</code>	Make alice group admin
<code>newgrp devops</code>	Switch primary group in current shell
<code>groups alice</code>	List groups for alice
<code>getent group devops</code>	Query group via NSS (works with LDAP too)

1.5 sudo & /etc/sudoers

sudo allows controlled privilege escalation. Always edit with `visudo` to prevent syntax errors.

```
# /etc/sudoers (edit with: visudo)
root ALL=(ALL:ALL) ALL
alice ALL=(ALL) NOPASSWD: /usr/bin/apt, /sbin/reboot
%wheel ALL=(ALL) ALL # all members of wheel group
%devops ALL=(ALL) NOPASSWD: ALL

# Drop-in files in /etc/sudoers.d/
# sudo -l # list alice's sudo privileges
# sudo -u bob command # run command as user bob
```

■ *TIP: Always use `visudo` — it validates syntax before saving, preventing lockouts.*

1.6 PAM — Pluggable Authentication Modules

PAM provides a flexible authentication framework. Configuration files live in `/etc/pam.d/`.

- Four module types: auth, account, password, session
- Control flags: required, requisite, sufficient, optional
- `pam_pwquality.so` — enforces password complexity

- pam_tally2.so / pam_faillock.so — account lockout after failed attempts
- pam_limits.so — resource limits from /etc/security/limits.conf

```
# /etc/pam.d/system-auth snippet
auth required pam_faillock.so preauth silent deny=5 unlock_time=600
auth sufficient pam_unix.so nullok try_first_pass
auth [default=die] pam_faillock.so authfail

# /etc/security/limits.conf
alice hard nproc 100 # limit processes
@devops soft nofile 4096 # open file limit for group
```

Chapter 2

File Permissions & ACL

chmod, SUID/SGID, umask, Access Control Lists

2.1 Standard Permission Bits

Every file/directory has a 10-character permission string:

```
- rwxr-xr-x -  
^  ^^^  ^^^  ^^^  
| owner group others  
type: - file d dir l symlink c char b block p pipe s socket
```

Command / Syntax	Description
chmod 755 file	rxwx-rx-x (numeric / octal)
chmod u=rwx,g=rx,o=r file	Symbolic mode
chmod +x script.sh	Add execute for all
chmod -w file	Remove write for all
chmod o-rwx dir	Remove all permissions for others
chmod -R 750 /var/app	Recursive
chown alice file	Change owner
chown alice:devops file	Change owner and group
chown -R alice:alice /home/alice	Recursive ownership change
chgrp devops file	Change group only
stat file	Show detailed permissions & timestamps
ls -la	Long listing with hidden files

2.2 Special Permission Bits

Bit	Octal	On File	On Directory
SUID	4000	Executes as owner (e.g. /usr/bin/passwd)	No effect
SGID	2000	Executes as group owner	New files inherit directory's group
Sticky	1000	No effect	Only owner/root can delete files inside

```
chmod 4755 /usr/bin/myprog # SUID  
chmod 2775 /var/shared # SGID on directory  
chmod 1777 /tmp # Sticky bit (like /tmp)  
# In ls output: s replaces x for SUID/SGID; t replaces x for sticky
```

■ **WARNING: SUID on shell scripts is ignored by the kernel. Only applies to ELF binaries.**

2.3 umask

umask subtracts permissions from the default (666 for files, 777 for dirs):

```
umask # Show current umask (e.g. 0022)  
umask 027 # Set umask: files=640, dirs=750
```

```
# Default - umask = actual permissions
# files: 666 - 022 = 644 (rw-r--r--)
# dirs: 777 - 022 = 755 (rwxr-xr-x)
# Set persistently in /etc/profile or ~/.bashrc
```

2.4 Access Control Lists (ACL)

ACLs extend standard permissions, allowing per-user and per-group access rules on any file/directory. Requires filesystem mounted with the 'acl' option (ext4/xfs have it by default on modern distros).

Command / Syntax	Description
getfacl file	View ACL entries
setfacl -m u:alice:rw file	Grant alice read/write
setfacl -m g:devops:rx dir	Grant devops group r-x
setfacl -m o::- file	Remove other permissions via ACL
setfacl -x u:alice file	Remove alice's ACL entry
setfacl -b file	Remove ALL ACL entries
setfacl -d -m g:devops:rwx dir	Set DEFAULT ACL (inherited by new files)
setfacl -R -m u:alice:rX dir	Recursive; X = execute only if dir/already exec
setfacl --restore=acl.bak	Restore ACLs from backup file
getfacl -R /data > acl.bak	Backup all ACLs recursively
cp --preserve=all src dst	Copy file preserving ACLs

```
# Example ACL output (getfacl /shared)
# file: /shared
# owner: root
# group: staff
user::rwx
user:alice:rw- # named user ACE
group::r-x
group:devops:rwx # named group ACE
mask::rwx # effective permission mask
other::r--
default:user::rwx # default ACEs for new files
default:group::r-x
default:other::---
```

■ *TIP: The 'mask' entry limits effective permissions for named users/groups. If mask is r--, even alice:rwx becomes r-- effective.*

2.5 Extended Attributes (xattr)

- getfattr -d file — list extended attributes
- setfattr -n user.comment -v 'my note' file
- chattr +i file — make file immutable (not even root can delete)
- chattr +a file — append-only (log files)
- lsattr file — list chattr flags

Chapter 3

LVM & Filesystem Management

Logical Volume Manager, Filesystems, Quota, Swap

3.1 LVM Architecture

LVM adds an abstraction layer between physical disks and filesystems, enabling flexible resizing, snapshots, and striping.

Layer	Object	Description
Physical Volume (PV)	/dev/sdb, /dev/sdc	Raw disk or partition initialised for LVM
Volume Group (VG)	vg_data	Pool of PVs; space divided into 4 MiB PEs by default
Logical Volume (LV)	/dev/vg_data/lv_home	Virtual partition formatted with a filesystem

PV Commands

Command / Syntax	Description
<code>pvcreate /dev/sdb /dev/sdc</code>	Initialise PVs
<code>pvdisplay</code>	Detailed PV info
<code>pvs</code>	Brief PV summary
<code>pvremove /dev/sdb</code>	Remove PV (VG must not use it)
<code>pvmove /dev/sdb</code>	Migrate data off PV

VG Commands

Command / Syntax	Description
<code>vgcreate vg_data /dev/sdb /dev/sdc</code>	Create VG from two PVs
<code>vgextend vg_data /dev/sdd</code>	Add PV to existing VG
<code>vgreduce vg_data /dev/sdc</code>	Remove PV from VG
<code>vgdisplay vg_data</code>	Detailed VG info
<code>vgs</code>	Brief VG summary
<code>vgrename old_name new_name</code>	Rename VG
<code>vgremove vg_data</code>	Delete VG (all LVs must be removed first)

LV Commands

Command / Syntax	Description
<code>lvcreate -L 20G -n lv_home vg_data</code>	Create 20 GiB LV
<code>lvcreate -l 100%FREE -n lv_data vg_data</code>	Use all free space
<code>lvcreate -l 50%VG -n lv_db vg_data</code>	Use 50% of VG
<code>lvextend -L +10G /dev/vg_data/lv_home</code>	Extend LV by 10 GiB
<code>lvextend -l +100%FREE /dev/vg_data/lv_home</code>	Extend using all free PE

Command / Syntax	Description
<code>lvreduce -L -5G /dev/vg_data/lv_home</code>	Shrink LV (unmount first!)
<code>lvresize -L 30G /dev/vg_data/lv_home</code>	Resize to exact size
<code>lvremove /dev/vg_data/lv_home</code>	Delete LV
<code>lvdisplay</code>	Detailed LV info
<code>lvs</code>	Brief LV summary

LVM Snapshots

```
lvcreate -L 5G -s -n snap_home /dev/vg_data/lv_home
# Mount snapshot (read-only)
mount -o ro /dev/vg_data/snap_home /mnt/snap
# Restore from snapshot
umount /dev/vg_data/lv_home
lvconvert --merge /dev/vg_data/snap_home
```

3.2 Filesystems

Filesystem	mkfs command	Notes
ext4	<code>mkfs.ext4 /dev/vg/lv</code>	Journalized; default on many distros; tune2fs for tuning
xfs	<code>mkfs.xfs /dev/vg/lv</code>	High performance; cannot shrink online; xfs_repair
btrfs	<code>mkfs.btrfs /dev/vg/lv</code>	CoW, built-in snapshots, RAID
vfat	<code>mkfs.vfat /dev/sdb1</code>	FAT32; cross-platform; no permissions
swap	<code>mkswap /dev/vg/lv_swap</code>	Not mounted like regular fs

Resize filesystem after LV extend

```
# ext4
resize2fs /dev/vg_data/lv_home # no unmount needed if online
# Or in one step:
lvextend -L +10G -r /dev/vg_data/lv_home # -r resizes fs automatically

# xfs (can only grow, not shrink)
xfs_growfs /mountpoint

# Shrinking ext4 (MUST unmount first)
umount /mountpoint
e2fsck -f /dev/vg_data/lv_home
resize2fs /dev/vg_data/lv_home 15G
lvreduce -L 15G /dev/vg_data/lv_home
mount /dev/vg_data/lv_home /mountpoint
```

■ **WARNING:** Always run `e2fsck` before shrinking an ext4 filesystem. XFS cannot be shrunk.

3.3 Mounting & /etc/fstab

```
# Manual mount
mount /dev/vg_data/lv_home /home
mount -t xfs -o noatime,nodiratime /dev/sdb1 /data
umount /home

# /etc/fstab format:
```

```
# device mountpoint fstype options dump pass
UUID=abc123... /home ext4 defaults,noatime 0 2
/dev/vg/lv_data /data xfs defaults 0 2
tmpfs /tmp tmpfs size=512M 0 0

# Get UUID
blkid /dev/sdb1
lsblk -f

mount -a # mount all /etc/fstab entries
findmnt # show mounted filesystems as tree
df -hT # disk usage with filesystem type
du -sh /home/* # directory sizes
```

3.4 Swap Management

```
mkswap /dev/vg_data/lv_swap
swapon /dev/vg_data/lv_swap
swapoff /dev/vg_data/lv_swap
swapon --show # show active swap
free -h # show RAM and swap usage

# Swapfile creation
dd if=/dev/zero of=/swapfile bs=1M count=2048
chmod 600 /swapfile
mkswap /swapfile
swapon /swapfile
# Add to fstab: /swapfile none swap sw 0 0
```

3.5 Disk Quotas

```
# 1. Add usrquota,grpquota to fstab options, then remount
# 2. Generate quota database
quotacheck -cugm /home
# 3. Enable quotas
quotaon -v /home
# 4. Set limits for user
edquota -u alice # opens quota in editor
# Or non-interactively:
setquota -u alice 500M 600M 0 0 /home # soft hard inodes-soft inodes-hard
# 5. View
quota -u alice
repquota -aus /home
```

Chapter 4

SELinux

Security-Enhanced Linux — Mandatory Access Control

SELinux implements Mandatory Access Control (MAC) on top of standard DAC. Every process and object has a security context, and policy rules decide what interactions are allowed.

4.1 SELinux Modes

Mode	Behaviour
Enforcing	Policy is enforced; denials are logged AND blocked
Permissive	Policy is not enforced; only logged — useful for troubleshooting
Disabled	SELinux is completely off — requires reboot to re-enable

```
getenforce # show current mode
setenforce 1 # set Enforcing (temporary)
setenforce 0 # set Permissive (temporary)

# Persistent: /etc/selinux/config
SELINUX=enforcing # enforcing | permissive | disabled
SELINUXTYPE=targeted # targeted (default) | mls | minimum
```

4.2 Security Contexts (Labels)

Format: user:role:type:level (e.g. system_u:object_r:httpd_sys_content_t:s0)

```
ls -Z /var/www/html/index.html
# unconfined_u:object_r:httpd_sys_content_t:s0

ps -eZ | grep httpd
# system_u:system_r:httpd_t:s0 1234 httpd

id -Z # show your SELinux context
```

- user — SELinux user (system_u, unconfined_u, staff_u)
- role — what roles the user can enter (system_r, unconfined_r)
- type — the most important part; policy rules are type:type allow rules
- level — MLS/MCS sensitivity level (s0 is default in targeted policy)

4.3 File Context Management

Command / Syntax	Description
chcon -t httpd_sys_content_t /var/www/html/index.html	Change context temporarily
restorecon -v /var/www/html/file	Restore default context
restorecon -Rv /var/www/html/	Recursive restore
semanage fcontext -l	List all default file contexts
semanage fcontext -a -t httpd_sys_content_t '/data(/.*)?'	Add persistent rule
semanage fcontext -d '/data(/.*)?'	Delete rule

Command / Syntax	Description
<code>matchpathcon /var/www/html</code>	Show default context for path

■ *TIP: After adding a custom fcontext rule with semanage, always run restorecon to apply it. chcon changes are temporary and lost on relabelling.*

4.4 SELinux Booleans

Booleans are on/off switches that adjust policy without custom modules:

Command / Syntax	Description
<code>getsebool -a</code>	List all booleans
<code>getsebool httpd_can_network_connect</code>	Show specific boolean
<code>setsebool httpd_can_network_connect on</code>	Enable (temporary)
<code>setsebool -P httpd_can_network_connect on</code>	Enable persistent
<code>semanage boolean -l</code>	List with descriptions

```
# Common booleans
httpd_can_network_connect # Apache can connect to network
httpd_enable_homedirs # Apache can serve ~/public_html
ftp_home_dir # FTP users can access home dirs
samba_enable_home_dirs # Samba can share home dirs
allow_ftp_anon_write # Anonymous FTP can write
```

4.5 Port Contexts

```
semanage port -l | grep http # show http-related ports
semanage port -a -t http_port_t -p tcp 8080 # allow Apache on 8080
semanage port -d -t http_port_t -p tcp 8080 # remove
```

4.6 Troubleshooting SELinux Denials

```
# View recent denials
ausearch -m avc -ts recent
ausearch -m avc -ts today | audit2why

# Generate policy module from denials
ausearch -m avc | audit2allow -m mymodule > mymodule.te
checkmodule -M -m mymodule.te -o mymodule.mod
semodule_package -m mymodule.mod -o mymodule.pp
semodule -i mymodule.pp

# Or directly (simpler but less targeted)
ausearch -m avc | audit2allow -M mymod && semodule -i mymod.pp

# Temporarily put a single process domain to permissive
semanage permissive -a httpd_t
semanage permissive -d httpd_t # remove
```

■ *NOTE: If SELinux is blocking a service, check /var/log/audit/audit.log and use setroubleshoot-server (sealert) for human-readable explanations.*

Chapter 5

DNS — Domain Name System

BIND, Zones, Records, Query Tools

5.1 DNS Concepts

- DNS maps hostnames ↔ IP addresses in a hierarchical, distributed database
- Resolvers query recursive resolvers → root servers → TLD servers → authoritative servers
- TTL (Time To Live) controls how long records are cached
- BIND (Berkeley Internet Name Domain) is the most common DNS server software

Record Types

Record	Purpose	Example
A	Hostname → IPv4	www IN A 192.168.1.10
AAAA	Hostname → IPv6	www IN AAAA 2001:db8::1
CNAME	Alias to another hostname	mail IN CNAME smtp.example.com.
MX	Mail exchange server	@ IN MX 10 mail.example.com.
PTR	IP → Hostname (reverse)	10.1.168.192.in-addr.arpa. IN PTR www.example.com.
NS	Authoritative nameserver	@ IN NS ns1.example.com.
SOA	Start of Authority (zone meta)	Contains serial, refresh, retry, expire
TXT	Arbitrary text (SPF, DKIM)	@ IN TXT 'v=spf1 mx ~all'
SRV	Service location	_http._tcp IN SRV 0 5 80 www.example.com.

5.2 BIND Configuration

```
# /etc/named.conf (RHEL/CentOS) or /etc/bind/named.conf (Debian/Ubuntu)
options {
  directory "/var/named";
  listen-on { 127.0.0.1; 192.168.1.0/24; };
  allow-query { localhost; 192.168.1.0/24; };
  recursion yes;
  forwarders { 8.8.8.8; 8.8.4.4; };
};

zone "example.com" IN {
  type master;
  file "example.com.zone";
  allow-update { none; };
};

zone "1.168.192.in-addr.arpa" IN {
  type master;
  file "192.168.1.rev";
};
```

Forward Zone File (/var/named/example.com.zone)

```
$TTL 86400
@ IN SOA ns1.example.com. admin.example.com. (
2024010101 ; Serial (YYYYMMDDnn)
3600 ; Refresh
1800 ; Retry
604800 ; Expire
86400 ) ; Negative TTL

@ IN NS ns1.example.com.
@ IN NS ns2.example.com.
@ IN MX 10 mail.example.com.
ns1 IN A 192.168.1.1
ns2 IN A 192.168.1.2
www IN A 192.168.1.10
mail IN A 192.168.1.20
ftp IN CNAME www
@ IN TXT 'v=spf1 mx ~all'
```

Reverse Zone File (/var/named/192.168.1.rev)

```
$TTL 86400
@ IN SOA ns1.example.com. admin.example.com. (
2024010101 3600 1800 604800 86400 )
@ IN NS ns1.example.com.
1 IN PTR ns1.example.com.
10 IN PTR www.example.com.
20 IN PTR mail.example.com.
```

Managing BIND

Command / Syntax	Description
named-checkconf	Validate named.conf syntax
named-checkzone example.com /var/named/example.com.zone	Validate zone file
systemctl restart named	Restart BIND
rndc reload	Reload zones without restart
rndc reload example.com	Reload specific zone
rndc status	Show BIND status
rndc flush	Flush DNS cache

5.3 DNS Query Tools

Command / Syntax	Description
dig example.com	Basic A record query
dig example.com MX	Query MX records
dig @8.8.8.8 example.com	Query specific DNS server
dig -x 192.168.1.10	Reverse lookup (PTR)
dig +short example.com	Short output
dig +trace example.com	Trace full resolution path
nslookup example.com	Interactive/non-interactive lookups

Command / Syntax	Description
host example.com	Simple forward/reverse lookup
host -t MX example.com	Query specific record type
systemd-resolve --status	Show systemd-resolved info

Chapter 6

Virtual Hosting

Apache & Nginx — Hosting Multiple Sites

6.1 Apache Virtual Hosts

Apache (httpd) supports name-based virtual hosting on the same IP:port — the Host: header decides which site is served.

Name-based VirtualHost (HTTP)

```
# /etc/apache2/sites-available/example.com.conf (Debian)
# /etc/httpd/conf.d/example.com.conf (RHEL)

ServerName example.com
ServerAlias www.example.com
DocumentRoot /var/www/example.com
ErrorLog /var/log/apache2/example.com_error.log
CustomLog /var/log/apache2/example.com_access.log combined

AllowOverride All
Require all granted

# Enable site (Debian/Ubuntu)
a2ensite example.com.conf
a2dissite example.com.conf
apache2ctl configtest
systemctl reload apache2

# RHEL/CentOS – just place file in /etc/httpd/conf.d/
httpd -t # syntax check
systemctl reload httpd
```

TLS/HTTPS with mod_ssl

```
# Enable modules (Debian)
a2enmod ssl rewrite headers

ServerName example.com
DocumentRoot /var/www/example.com
SSLEngine on
SSLCertificateFile /etc/ssl/certs/example.com.crt
SSLCertificateKeyFile /etc/ssl/private/example.com.key
SSLCertificateChainFile /etc/ssl/certs/ca-chain.crt
SSLProtocol all -SSLv3 -TLSv1 -TLSv1.1
SSLCipherSuite HIGH:!aNULL:!MD5
Header always set Strict-Transport-Security 'max-age=31536000'

# HTTP → HTTPS redirect in the port 80 VirtualHost
RewriteEngine on
RewriteRule ^ https://%{HTTP_HOST}%{REQUEST_URI} [R=301,L]
```

Useful Apache Directives

Command / Syntax	Description
ServerName / ServerAlias	Primary and alternate names
DocumentRoot	Root directory for web files
DirectoryIndex	Default file to serve (index.html)
Alias /path /fs/path	Map URL to different directory
ProxyPass / ProxyPassReverse	Reverse proxy to backend
AuthType Basic	HTTP Basic Authentication
Require valid-user	Require authentication
Options -Indexes	Disable directory listing
LimitRequestBody 10485760	Limit upload size to 10 MiB
LogLevel warn	Logging verbosity

6.2 Nginx Virtual Hosts

```
# /etc/nginx/sites-available/example.com (Debian)
# /etc/nginx/conf.d/example.com.conf (RHEL)
server {
    listen 80;
    server_name example.com www.example.com;
    root /var/www/example.com;
    index index.html index.php;

    location / {
        try_files $uri $uri/ =404;
    }

    location ~ \.php$ {
        include fastcgi_params;
        fastcgi_pass unix:/run/php/php8.1-fpm.sock;
        fastcgi_param SCRIPT_FILENAME $document_root$fastcgi_script_name;
    }

    access_log /var/log/nginx/example.com_access.log;
    error_log /var/log/nginx/example.com_error.log;
}

# HTTPS block
server {
    listen 443 ssl http2;
    server_name example.com;
    ssl_certificate /etc/ssl/certs/example.com.crt;
    ssl_certificate_key /etc/ssl/private/example.com.key;
    ssl_protocols TLSv1.2 TLSv1.3;
}

# Enable (Debian)
ln -s /etc/nginx/sites-available/example.com /etc/nginx/sites-enabled/
nginx -t && systemctl reload nginx
```

6.3 Let's Encrypt (Certbot)

```
# Install certbot
apt install certbot python3-certbot-apache # Debian
dnf install certbot python3-certbot-apache # RHEL

# Obtain and install cert automatically
certbot --apache -d example.com -d www.example.com
certbot --nginx -d example.com

# Renew all certs
certbot renew --dry-run
certbot renew

# Auto-renew via systemd timer (usually set up by certbot)
systemctl status certbot.timer
```

Chapter 7

Mail Server — Postfix & Dovecot

MTA, SMTP AUTH, IMAP/POP3, TLS

7.1 Mail Flow Architecture

- MUA (Mail User Agent) — client (Thunderbird, Outlook)
- MTA (Mail Transfer Agent) — Postfix receives/relays/delivers SMTP
- MDA (Mail Delivery Agent) — delivers to local mailbox (Dovecot's LDA or Procmail)
- MRA (Mail Retrieval Agent) — Dovecot serves IMAP/POP3 to MUAs
- Standard ports: SMTP=25, Submission=587, SMTPS=465, IMAP=143, IMAPS=993, POP3=110, POP3S=995

7.2 Postfix — main.cf Key Parameters

```
# /etc/postfix/main.cf
myhostname = mail.example.com
mydomain = example.com
myorigin = $mydomain
inet_interfaces = all
inet_protocols = all
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
mynetworks = 127.0.0.0/8, 192.168.1.0/24
home_mailbox = Maildir/
mailbox_size_limit = 0
message_size_limit = 52428800 # 50 MiB
smtpd_banner = $myhostname ESMTP

# Relay (if using smarthost)
relayhost = [smtp.isp.com]:587
```

Postfix TLS (Inbound & Outbound)

```
# Inbound TLS (smtpd)
smtpd_tls_cert_file = /etc/ssl/certs/mail.crt
smtpd_tls_key_file = /etc/ssl/private/mail.key
smtpd_tls_security_level = may # opportunistic
smtpd_tls_loglevel = 1

# Outbound TLS (smtp)
smtp_tls_security_level = may
smtp_tls_loglevel = 1
smtp_tls_CAfile = /etc/ssl/certs/ca-certificates.crt
```

SMTP Authentication with SASL

```
# main.cf additions
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth
smtpd_sasl_auth_enable = yes
smtpd_sasl_security_options = noanonymous
smtpd_recipient_restrictions =
permit_mynetworks,
permit_sasl_authenticated,
reject_unauth_destination
```

```
# master.cf – enable submission port 587
submission inet n - y - - smtpd
-o syslog_name=postfix/submission
-o smtpd_tls_security_level=encrypt
-o smtpd_sasl_auth_enable=yes
-o smtpd_client_restrictions=permit_sasl_authenticated,reject
```

Postfix Management Commands

Command / Syntax	Description
postfix check	Validate configuration
postfix reload	Reload config without restart
postmap /etc/postfix/virtual	Compile lookup table
postqueue -p	View mail queue
postqueue -f	Flush (retry) mail queue
postsuper -d ALL	Delete all queued messages
postsuper -d <id>	Delete specific message
postcat -q <id>	View queued message
mailq	Alias for postqueue -p
sendmail -v alice@example.com	Send test mail interactively

7.3 Dovecot — IMAP/POP3 Server

```
# /etc/dovecot/dovecot.conf
protocols = imap pop3 lmtp

# /etc/dovecot/conf.d/10-mail.conf
mail_location = maildir:~/Maildir

# /etc/dovecot/conf.d/10-auth.conf
auth_mechanisms = plain login
disable_plaintext_auth = yes # require SSL

# /etc/dovecot/conf.d/10-ssl.conf
ssl = required
ssl_cert = ssl_key = ssl_min_protocol = TLSv1.2

# /etc/dovecot/conf.d/10-master.conf – SASL socket for Postfix
service auth {
  unix_listener /var/spool/postfix/private/auth {
    mode = 0660
    user = postfix
    group = postfix
  }
}
```

Testing Mail Configuration

```
# Test SMTP connection
telnet mail.example.com 25
EHLO test.local
```

```
QUIT

# Test with openssl (TLS)
openssl s_client -connect mail.example.com:587 -starttls smtp
openssl s_client -connect mail.example.com:993 # IMAPS

# Send test mail via swaks
swaks --to alice@example.com --from bob@example.com \
--server mail.example.com --auth --tls

# Check logs
tail -f /var/log/mail.log # Debian
journalctl -fu postfix # systemd
```

Chapter 8

FTP — File Transfer Protocol

vsftpd, ProFTPD, Active/Passive, FTPS

8.1 Active vs Passive Mode

	Active Mode	Passive Mode
Control channel	Client → Server:21	Client → Server:21
Data channel	Server → Client (random port)	Client → Server (PASV port, e.g. 50000-51000)
Firewall friendliness	Poor (client firewall blocks incoming)	Good (client initiates data connection)
Use case	Legacy / internal networks	Internet / NAT / modern use

8.2 vsftpd Configuration

```
# /etc/vsftpd.conf
listen=YES
listen_ipv6=NO
anonymous_enable=NO # disable anonymous login
local_enable=YES # allow system users
write_enable=YES
local_umask=022
chroot_local_user=YES # jail users in home dir
allow_writeable_chroot=YES # needed if home is writeable
secure_chroot_dir=/var/run/vsftpd/empty

# Passive mode
pasv_enable=YES
pasv_min_port=50000
pasv_max_port=51000
pasv_address=203.0.113.10 # public IP (NAT scenarios)

# FTP over TLS (FTPS)
ssl_enable=YES
rsa_cert_file=/etc/ssl/certs/vsftpd.crt
rsa_private_key_file=/etc/ssl/private/vsftpd.key
force_local_logins_ssl=YES
force_local_data_ssl=YES
ssl_tlsv1_2=YES

# Restrict users
userlist_enable=YES
userlist_file=/etc/vsftpd.userlist
userlist_deny=NO # ALLOW only users in the list

# Logging
xferlog_enable=YES
xferlog_file=/var/log/vsftpd.log
```

8.3 Anonymous FTP

```
anonymous_enable=YES
anon_root=/var/ftp/pub
no_anon_password=YES # no password prompt
anon_upload_enable=NO # disable uploads
anon_mkdir_write_enable=NO

# Directory permissions for anonymous
# /var/ftp must be owned by root with no write for others
chown root:root /var/ftp
chmod 755 /var/ftp
chmod 777 /var/ftp/incoming # if uploads allowed
```

8.4 FTP Command Reference

Command / Syntax	Description
ftp hostname	Connect (plain FTP)
lftp -u user,pass hostname	Feature-rich CLI FTP client
ncftp hostname	Persistent sessions
curl ftp://host/file -u user:pass	Download via curl
curl -T file ftp://host/ -u user:pass	Upload via curl
wget ftp://user:pass@host/file	Download via wget
sftp user@host	SSH-based secure FTP (not FTP over TLS)
scp file user@host:/path	Secure copy over SSH

■ *TIP: Prefer SFTP (SSH File Transfer Protocol) or FTPS over plain FTP. Plain FTP sends credentials in cleartext.*

Chapter 9

Samba / CIFS

Windows File & Print Sharing from Linux

9.1 Samba Overview

- Samba implements SMB/CIFS protocol for interoperability with Windows
- Can act as: file server, print server, or Active Directory Domain Controller
- Key daemons: smbd (file/print), nmbd (NetBIOS name resolution), winbindd (AD integration)
- Config file: /etc/samba/smb.conf

9.2 smb.conf Structure

```
# /etc/samba/smb.conf
[global]
workgroup = WORKGROUP
server string = Samba File Server
netbios name = LINUXSERVER
security = user # user-level auth
map to guest = Bad User # unknown users → guest
log file = /var/log/samba/log.%m
max log size = 50
passdb backend = tdbsam
printing = cups
printcap name = cups

[homes]
comment = Home Directories
browseable = no
writable = yes
valid users = %S

[public]
comment = Public Share
path = /srv/samba/public
browseable = yes
read only = no
guest ok = yes
create mask = 0664
directory mask = 0775

[devops]
comment = DevOps Team Share
path = /srv/samba/devops
valid users = @devops
writable = yes
create mask = 0770
directory mask = 0770
```

9.3 User & Password Management

Command / Syntax	Description
<code>smbpasswd -a alice</code>	Add alice to Samba password database
<code>smbpasswd -e alice</code>	Enable Samba account
<code>smbpasswd -d alice</code>	Disable Samba account
<code>smbpasswd -x alice</code>	Delete Samba account
<code>pdbedit -L</code>	List Samba users
<code>pdbedit -Lv alice</code>	Detailed user info
<code>testparm</code>	Validate smb.conf syntax
<code>testparm -s</code>	Show effective config
<code>smbstatus</code>	Show active connections
<code>net usershare list</code>	List user shares

■ **NOTE:** Samba maintains a separate password database. System users must be added to Samba with `smbpasswd -a`. Their system password and Samba password are independent.

9.4 Mounting CIFS Shares from Linux

```
# Manual mount
mount -t cifs //server/share /mnt/share \
-o username=alice,password=secret,workgroup=WORKGROUP

# Better: use credentials file
echo 'username=alice' > /etc/samba/.creds
echo 'password=secret' >> /etc/samba/.creds
chmod 600 /etc/samba/.creds
mount -t cifs //server/share /mnt/share -o credentials=/etc/samba/.creds

# /etc/fstab entry
//server/share /mnt/share cifs
credentials=/etc/samba/.creds,uid=1000,gid=1000,iocharset=utf8 0 0

# Query shares on a server
smbclient -L //server -U alice

# Interactive shell
smbclient //server/share -U alice
smb: \> ls
smb: \> get file.txt
smb: \> put upload.txt
```

9.5 SELinux & Firewall for Samba

```
# SELinux booleans
setsebool -P samba_enable_home_dirs on
setsebool -P samba_export_all_rw on
chcon -t samba_share_t /srv/samba/devops # or use semanage fcontext

# Firewall
firewall-cmd --permanent --add-service=samba
firewall-cmd --reload

# Or with ufw:
ufw allow samba
```

Chapter

10

IMAP & POP3 Deep Dive

Protocols, Dovecot Config, Mailbox Formats

10.1 IMAP vs POP3

Feature	IMAP	POP3
Storage	Mail stays on server	Mail downloaded & deleted from server
Multi-device	Full sync across devices	Not designed for multi-device
Folders	Server-side folders	Single inbox only
Port	143 (STARTTLS), 993 (SSL)	110 (STARTTLS), 995 (SSL)
State	Stateful (flags: read,answered)	Stateless
Use case	Modern email clients	Legacy / bandwidth-limited

10.2 Mailbox Formats

- mbox — single file per mailbox, all messages concatenated; simple but slow with locks
- Maildir — one file per message in a directory tree (cur/, new/, tmp/); lockless, fast, used by Dovecot
- dbox (Dovecot) — Dovecot's own format, more efficient for IMAP

```
# Maildir structure for user alice
~/Maildir/
new/ <- new unread messages
cur/ <- messages that have been seen
tmp/ <- messages being written (transient)
.Sent/new/ .Drafts/new/ .Trash/new/ <- sub-folders
```

10.3 Dovecot Configuration Reference

```
# /etc/dovecot/conf.d/20-imap.conf
protocol imap {
mail_plugins = $mail_plugins imap_quota
imap_idle_notify_interval = 2 mins
}

# /etc/dovecot/conf.d/20-pop3.conf
protocol pop3 {
pop3_uidl_format = %08Xu%08Xv
}

# /etc/dovecot/conf.d/15-lda.conf
protocol lda {
mail_plugins = sieve # server-side filtering
}

# Quota plugin - /etc/dovecot/conf.d/90-quota.conf
plugin {
quota = maildir:User quota
quota_rule = *:storage=1G
```

```
quota_rule2 = Trash:storage=100M
}
```

Dovecot Management

Command / Syntax	Description
doveadm user alice	Show user's mailbox info
doveadm quota get -u alice	Show alice's quota usage
doveadm quota recalc -u alice	Recalculate quota
doveadm expunge -u alice mailbox INBOX before 30d	Delete old mail
doveadm search -u alice mailbox INBOX ALL	Search alice's inbox
doveadm purge -u alice	Expunge flagged messages
doveadm reload	Reload Dovecot config
dovecot -n	Show non-default settings

Testing IMAP Manually

```
# IMAPS with openssl
openssl s_client -connect mail.example.com:993
# After connection (Dovecot prompt: * OK Dovecot ready)
A001 LOGIN alice mypassword
A002 LIST '' '*'
A003 SELECT INBOX
A004 FETCH 1 BODY[]
A005 LOGOUT
```

Chapter

11

Networking Essentials

ip, ss, iptables, nftables, firewallld, SSH

11.1 Network Configuration Commands

Command / Syntax	Description
ip addr show	Show interfaces and IPs
ip addr add 192.168.1.10/24 dev eth0	Assign IP
ip addr del 192.168.1.10/24 dev eth0	Remove IP
ip link set eth0 up/down	Bring interface up or down
ip route show	Show routing table
ip route add default via 192.168.1.1	Set default gateway
ip route add 10.0.0.0/8 via 192.168.1.254	Add static route
ip neigh show	Show ARP/NDP table
ss -tulnp	TCP/UDP listening sockets with PIDs
ss -s	Socket statistics summary
netstat -tulnp	Legacy equivalent of ss -tulnp
nmcli con show	List NetworkManager connections
nmcli con up 'Wired'	Bring up NM connection
nmcli dev status	Show device status
nmtui	Text UI for NetworkManager
ping -c 4 8.8.8.8	ICMP connectivity test
traceroute / tracepath	Show packet path to destination
mtr host	Combined ping+traceroute
curl -I https://example.com	HTTP header test
nc -zv host 443	TCP port connectivity test

11.2 Firewall — iptables

```
# View rules
iptables -L -nv --line-numbers
iptables -t nat -L -nv

# Allow established/related
iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT

# Allow SSH, HTTP, HTTPS
iptables -A INPUT -p tcp --dport 22 -j ACCEPT
iptables -A INPUT -p tcp -m multiport --dports 80,443 -j ACCEPT

# Drop all other input
```

```
iptables -P INPUT DROP
iptables -P FORWARD DROP
iptables -P OUTPUT ACCEPT

# NAT (masquerade outbound)
iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
echo 1 > /proc/sys/net/ipv4/ip_forward

# Save / restore
iptables-save > /etc/iptables/rules.v4
iptables-restore < /etc/iptables/rules.v4
```

11.3 firewalld

Command / Syntax	Description
firewall-cmd --state	Check firewalld status
firewall-cmd --get-active-zones	Show active zones
firewall-cmd --zone=public --list-all	Show zone rules
firewall-cmd --permanent --add-service=http	Allow HTTP permanently
firewall-cmd --permanent --add-port=8080/tcp	Allow custom port
firewall-cmd --permanent --remove-service=ftp	Remove service
firewall-cmd --reload	Apply permanent rules
firewall-cmd --permanent --add-rich-rule='rule family=ipv4 source address=10.0.0.0/8 accept'	Allow subnet
firewall-cmd --permanent --add-masquerade	Enable NAT masquerade
firewall-cmd --zone=trusted --add-interface=eth0	Move interface to trusted zone

11.4 SSH Hardening

```
# /etc/ssh/sshd_config hardening
Port 2222 # non-standard port
PermitRootLogin no
PasswordAuthentication no # key-only auth
PubkeyAuthentication yes
AuthorizedKeysFile .ssh/authorized_keys
MaxAuthTries 3
LoginGraceTime 30
AllowUsers alice bob
AllowGroups sshusers
X11Forwarding no
UseDNS no
Protocol 2
ClientAliveInterval 300
ClientAliveCountMax 2

systemctl reload sshd

# Key management
ssh-keygen -t ed25519 -C 'alice@example.com'
ssh-copy-id -i ~/.ssh/id_ed25519.pub alice@server
ssh -i ~/.ssh/id_ed25519 alice@server
```

Chapter
12

Systemd & Service Management

systemctl, Unit Files, journalctl, Timers

12.1 systemctl Commands

Command / Syntax	Description
systemctl start service	Start a service
systemctl stop service	Stop a service
systemctl restart service	Restart
systemctl reload service	Reload config without restart
systemctl enable service	Enable at boot
systemctl disable service	Disable at boot
systemctl enable --now service	Enable and start immediately
systemctl status service	Show status & recent logs
systemctl is-active service	Returns active/inactive
systemctl is-enabled service	Returns enabled/disabled
systemctl list-units --type=service	List all services
systemctl list-units --failed	List failed units
systemctl daemon-reload	Reload unit files after changes
systemctl isolate multi-user.target	Switch to multi-user (no GUI)
systemctl get-default	Show default target
systemctl set-default graphical.target	Set default target
systemctl poweroff / reboot / halt	Power management

12.2 Unit File Anatomy

```
# /etc/systemd/system/myapp.service
[Unit]
Description=My Application Server
Documentation=https://example.com/docs
After=network.target postgresql.service
Requires=postgresql.service

[Service]
Type=simple # simple|forking|oneshot|notify|dbus
User=myapp
Group=myapp
WorkingDirectory=/opt/myapp
EnvironmentFile=/etc/myapp/env
ExecStartPre=/opt/myapp/pre-start.sh
ExecStart=/opt/myapp/server --port 8080
ExecReload=/bin/kill -HUP $MAINPID
Restart=on-failure
```

```
RestartSec=5
StandardOutput=journal
StandardError=journal
LimitNOFILE=65536

[Install]
WantedBy=multi-user.target
```

12.3 journalctl — Log Management

Command / Syntax	Description
journalctl	All logs (oldest first)
journalctl -f	Follow live (like tail -f)
journalctl -u nginx	Logs for nginx unit only
journalctl -u nginx -f	Follow nginx logs
journalctl --since '1 hour ago'	Logs from last hour
journalctl --since '2024-01-01' --until '2024-01-02'	Date range
journalctl -p err	Error-level and above
journalctl -p warning..crit	Range of priorities
journalctl --disk-usage	Show journal disk usage
journalctl --vacuum-size=500M	Trim journal to 500 MiB
journalctl --vacuum-time=30d	Remove entries older than 30 days
journalctl -b	Logs from current boot
journalctl -b -1	Logs from previous boot
journalctl -k	Kernel messages only

12.4 Systemd Timer Units

```
# /etc/systemd/system/backup.timer
[Unit]
Description=Daily backup timer

[Timer]
OnCalendar=daily
OnCalendar=Mon-Fri 03:00:00 # alternative
Persistent=true # run if missed
Unit=backup.service

[Install]
WantedBy=timers.target

# Enable and start the timer (not the service)
systemctl enable --now backup.timer
systemctl list-timers --all
```


**Chapter
13**

Package Management

apt, dnf/yum, rpm, zypper — All Distros

13.1 Debian/Ubuntu — apt & dpkg

Command / Syntax	Description
apt update	Refresh repository metadata
apt upgrade	Upgrade all packages
apt full-upgrade	Upgrade + resolve dependency changes
apt install nginx	Install package
apt install ./package.deb	Install local .deb
apt remove nginx	Remove (keep config)
apt purge nginx	Remove + delete config
apt autoremove	Remove unused dependencies
apt search nginx	Search for package
apt show nginx	Show package details
apt list --installed	List installed packages
apt-cache policy nginx	Show version & repo info
dpkg -i package.deb	Install .deb file
dpkg -l grep nginx	List matching packages
dpkg -L nginx	List files installed by package
dpkg -S /usr/sbin/nginx	Find package owning a file
dpkg-reconfigure tzdata	Reconfigure package

13.2 RHEL/CentOS/Fedora — dnf & rpm

Command / Syntax	Description
dnf check-update	Check for updates
dnf update	Update all packages
dnf install httpd	Install package
dnf install ./package.rpm	Install local .rpm
dnf remove httpd	Remove package
dnf autoremove	Remove unused deps
dnf search httpd	Search packages
dnf info httpd	Package details
dnf list installed	List installed
dnf provides /usr/sbin/httpd	Which package owns file

Command / Syntax	Description
<code>dnf history</code>	Transaction history
<code>dnf history undo 5</code>	Undo transaction #5
<code>dnf repolist</code>	List enabled repos
<code>dnf config-manager --enable repo</code>	Enable a repository
<code>rpm -ivh package.rpm</code>	Install RPM (verbose)
<code>rpm -qa grep httpd</code>	Query all — find httpd
<code>rpm -ql httpd</code>	List files in package
<code>rpm -qf /usr/sbin/httpd</code>	Which RPM owns file
<code>rpm -V httpd</code>	Verify package integrity
<code>rpm --import /etc/pki/rpm-gpg/RPM-GPG-KEY-import-GPG-key</code>	Import GPG key

13.3 openSUSE — zypper

Command / Syntax	Description
<code>zypper refresh</code>	Refresh repos
<code>zypper update</code>	Update all
<code>zypper install pkg</code>	Install
<code>zypper remove pkg</code>	Remove
<code>zypper search pkg</code>	Search
<code>zypper info pkg</code>	Package info
<code>zypper lr</code>	List repos

13.4 Compiling from Source

```
# Classic autotools build
tar xzf package-1.0.tar.gz && cd package-1.0
./configure --prefix=/usr/local --enable-feature
make -j$(nproc)
make install

# CMake build
mkdir build && cd build
cmake .. -DCMAKE_INSTALL_PREFIX=/usr/local
make -j$(nproc) && make install

# Install build dependencies first
apt build-dep nginx # Debian
dnf builddep nginx # RHEL
```

**Chapter
14**

Shell Scripting & Automation

Bash, cron, grep/sed/awk, find

14.1 Bash Scripting Essentials

```
#!/bin/bash
set -euo pipefail # exit on error, unset vars, pipe failures

# Variables
NAME='Alice'
AGE=30
echo "Hello, $NAME. You are $AGE years old."

# Conditionals
if [[ -f /etc/passwd ]]; then
echo 'File exists'
elif [[ -d /etc ]]; then
echo 'Directory exists'
else
echo 'Neither'
fi

# Arithmetic
RESULT=$(( 5 * 3 + 2 ))

# Loops
for f in /var/log/*.log; do
echo "Processing: $f"
done

COUNTER=0
while [[ $COUNTER -lt 5 ]]; do
echo $COUNTER
(( COUNTER++ ))
done

# Functions
backup_dir() {
local src=$1
local dst=$2
tar czf "${dst}/backup-$(date +%F).tar.gz" "$src"
}
backup_dir /etc /backup

# Arrays
SERVERS=(web1 web2 db1)
for s in "${SERVERS[@]}"; do ping -c1 $s; done
```

14.2 cron & at — Job Scheduling

```
crontab -e # edit current user's crontab
crontab -l # list crontab
crontab -r # remove crontab
```

```
# Cron format: minute hour day month weekday command
# * * * * * command
# | | | | |
# | | | | +-- day of week (0=Sun ... 7=Sun)
# | | | +----- month (1-12)
# | | +----- day of month (1-31)
# | +----- hour (0-23)
# +----- minute (0-59)

0 2 * * * /usr/bin/backup.sh # daily at 02:00
*/15 * * * * /usr/bin/check_disk.sh # every 15 minutes
0 0 1 * * /usr/bin/monthly_report.sh # 1st of each month
0 9-17 * * 1-5 /usr/bin/work_task.sh # Mon-Fri 9-17

# System cron files
# /etc/cron.daily/, /etc/cron.weekly/, /etc/cron.monthly/
# /etc/crontab – system crontab with user field

# at – run once at specific time
at 15:30 tomorrow
at> /usr/bin/my-script.sh
at>
atq # list pending jobs
atrm 3 # remove job #3
```

14.3 Text Processing — grep, sed, awk

Command / Syntax	Description
grep 'pattern' file	Basic search
grep -i 'error' /var/log/syslog	Case-insensitive
grep -r 'TODO' /opt/app/	Recursive search
grep -v '^#' /etc/ssh/sshd_config	Exclude comment lines
grep -c 'FAILED' auth.log	Count matches
grep -n 'error' file	Show line numbers
grep -E 'err warn crit' syslog	Extended regex (ERE)
grep -P '\d{4}-\d{2}-\d{2}' log	Perl-compatible regex
sed 's/old/new/g' file	Replace all occurrences
sed -i 's/foo/bar/g' file	In-place edit
sed -n '5,10p' file	Print lines 5-10
sed '/^#/d' file	Delete comment lines
awk '{print \$1,\$3}' file	Print columns 1 and 3
awk -F: '{print \$1}' /etc/passwd	Use : as delimiter
awk 'NR>5 && NR<10' file	Print lines 6-9
awk '\$3 > 100 {print \$1}' file	Conditional print
awk '{sum+=\$2} END{print sum}' file	Sum column 2

14.4 find & xargs

Command / Syntax	Description
<code>find /home -name '*.log'</code>	Find by name
<code>find /var -name '*.log' -mtime +30</code>	Files older than 30 days
<code>find /tmp -type f -size +100M</code>	Files over 100 MiB
<code>find / -perm -4000 -type f</code>	Find SUID files
<code>find /etc -user root -group root</code>	By owner
<code>find . -name '*.tmp' -delete</code>	Find and delete
<code>find . -name '*.log' xargs grep 'ERROR'</code>	Find then grep
<code>find . -type f xargs chmod 644</code>	Find then chmod
<code>find . -maxdepth 2 -type d</code>	Limit directory depth
<code>find . -newer /etc/passwd</code>	Newer than reference file

14.5 Useful One-liners

```
# List top 10 largest files
du -ah / 2>/dev/null | sort -rh | head -10

# Monitor log in real time filtering for errors
tail -f /var/log/syslog | grep --line-buffered -i 'error\|warn'

# Check open ports (no ss/netstat needed)
cat /proc/net/tcp

# Find users with empty passwords
awk -F: '($2 == "") {print $1}' /etc/shadow

# Tar + compress + pipe over SSH
tar czf - /data | ssh user@remote 'cat > /backup/data.tar.gz'

# Replace multiple spaces with single space
tr -s ' ' < file

# Count words per line and sort
awk '{print NF, $0}' file | sort -n
```

End of Linux System Administration Notes — LFCS Reference

Covers all major LFCS domains: Essential Commands, Operation of Running Systems, User and Group Management, Networking, Service Configuration, Storage Management.